Sai Uday R Udumula

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github.com/USUDR2604/ | SaiUdayR.Portfolio

SUMMARY

Data Engineer with 4+ years of experience building large-scale data platforms across telecom, cloud, and academia. Specialized in PySpark/Databricks pipelines processing 50M+ daily events, end-to-end ETL/ELT orchestration with Airflow and dbt, and multi-cloud deployments on AWS and GCP. Proven impact reducing pipeline latency by 30% and enabling real-time analytics at scale

EDUCATION

University of Missouri–Kansas City (UMKC) **Jan 2024 - Dec 2025**
Masters, Computer Science with Data Science *Kansas City, MO, USA*

EXPERIENCE

University of Missouri–Kansas City (UMKC)
Big Data & Operations Analyst – Data Engineering **Jan 2025 - Dec 2025**

- Designed 20+ Airflow ETL/ELT pipelines on Snowflake and Databricks, ingesting real-time university dining and facility operational data, reducing manual data preparation by 30% through automated ingestion, data validation checks, schema enforcement, and monitoring to ensure SLA-compliant, reliable data pipelines
- Analyzed large-scale operational datasets - food consumption, footfall, facility usage, and scheduling - using PySpark and SQL, improving resource allocation and schedule management accuracy by 15–20%.
- Built scalable data models using Spark, Airflow, and dbt enabling 15+ real-time Tableau dashboards delivering live scheduling insights and operational analysis, reducing reporting latency by 20%.
- **Skills Used:** Python, PySpark, SQL, Airflow, Snowflake, Databricks, dbt, Tableau, Parquet, JSON/CSV

Tata Consultancy Services (TCS)
Data Engineer – Big Data & Cloud Systems **Jul 2022 - Dec 2023**

- Built distributed PySpark pipelines on Databricks Lakehouse using Medallion Architecture (Bronze/Silver/Gold), processing 50M+ daily Vodacom network events and logs, reducing latency by 30% using Parquet/JSON formats, Delta Lake ACID transactions, and fault-tolerant distributed systems architecture.
- Engineered 10+ Airflow and dbt ETL/ELT pipelines on AWS S3 and GCP Data Lake, implementing CDC and SCD Type 1/2 patterns across Vodacom network datasets, improving pipeline reliability by 25% through automated ingestion and distributed compute workload optimization.
- Designed and optimized 50+ Snowflake data models to analyze Vodacom network performance, bandwidth, and recharge workloads, leveraging complex joins, window functions, and query optimization across 50M+ daily records; this enabled cost-optimization planning and improved network-capacity decision making
- Developed a full-stack Python Django backend and Angular/React frontend integrating SQL and NoSQL (MongoDB, Redis) databases into a unified service layer, enabling real-time Vodacom network data extraction, issue resolution workflows, and personalized server management, improving system response latency by 20%.
- Provisioned cloud infrastructure using Terraform and CloudFormation on AWS and GCP, deployed 10+ microservices with Docker and Kubernetes, and monitored server logs with the ELK Stack, Tableau, and Power BI, improving scalability by 30% and reducing incident detection time by 25–30%
- Architected a self-driven AI chatbot integrated with Big Data pipelines and RAG-based systems, enabling automated customer detail analysis, real-time query resolution on telecom account and network data, and reducing support load by 25%.
- **Skills Used:** Python, PySpark, Databricks Lakehouse, Delta Lake, Medallion Architecture, Airflow, dbt, Snowflake, ETL/ELT, CDC, SCD Type 1/2, Django, Angular, AWS (S3, EMR, Glue), GCP (Dataflow, BigQuery), Terraform, Docker, Kubernetes, ELK Stack, Tableau, Power BI, LangChain, LangGraph, RAG, Vector Databases, Agentic AI

Software Engineer Intern (Data Engineering & Big Data) **Feb 2022 - Jun 2022**

- Built an AI-powered assessment chatbot with RASA, LangChain, and LangGraph, using an Angular/React frontend and Python backend; the bot adapted difficulty, answered questions, explained errors, and generated personalized performance reports, which increased user completion rates and provided targeted improvement suggestions
- Built real-time data pipelines using PySpark and AWS to process user interaction and assessment data, enabling low-latency analytics under 3 seconds and scalable data processing with 25% improved feedback accuracy.
- Designed and managed Data Lake architecture on AWS and GCP using Snowflake, dbt, and Airflow, orchestrating ETL/ELT workflows across assessment and user interaction datasets, ensuring data quality, schema enforcement, and reliable downstream analytics with under 50ms retrieval latency via MongoDB and Redis caching.
- **Skills Used:** Python, RASA, LangChain, LangGraph, Agentic AI, RAG, Vector Databases, Deep Learning, Predictive Modeling, Prompt Engineering, PySpark, Apache Spark, MongoDB, SQL, Redis, AngularJS, ReactJS, AWS, GCP, Databricks Lakehouse.

TwoWaits
Software Engineer (Data Engineering & Backend Systems) **Jul 2020 - Dec 2021**

- Architected Big Data and microservices infrastructure using Apache Spark, Kafka, Airflow, and Kubernetes on AWS, building 10+ ETL/ELT pipelines orchestrating answer submission, validation, scoring, and report generation across Parquet/JSON/CSV sources with Snowflake and dbt, improving pipeline reliability by 25%.
- Built an NLP-driven educational chatbot using RASA and intent classification serving 2,000+ users, improving query resolution by 20%, delivering adaptive difficulty levels, doubt answering, incorrect answer explanations, and personalized performance reports through real-time conversational flows.
- Built real-time data pipelines using PySpark, MongoDB, and Redis caching on AWS to process user response data from submission to scoring, enabling low-latency analytics with sub-50ms feedback latency and improving system reliability by 25%.
- **Skills Used:** Python, RASA, NLP, RAG, Vector Databases, Deep Learning, Predictive Modeling, ETL/ELT, Apache Spark, Kafka, Airflow, MongoDB, Redis, Snowflake, dbt, AWS.

SKILLS

- **Data Engineering**: PySpark, Apache Spark , Databricks Lakehouse, Delta Lake, ACID Transactions, Medallion Architecture, Apache Airflow, ETL/ELT Pipelines, Data Pipeline Orchestation, Streaming & Batch Processing, Apache Kafka, Apache Flink, Snowflake, Data Warehousing, dbt, Data Modeling, Data Lakes, Data Lake Architecture, CDC, SCD Types, Schema Design, Schema Evolution, Data Lineage, Data Governance, Data Quality Monitoring, Cost Optimization, Query Optimization
- **Cloud Platforms**: AWS (S3, EMR, Glue, Lambda, Redshift), Google Cloud Platform (BigQuery, Dataflow), Azure (Data Lake, Databricks)
- **Machine Learning & Artificial Intelligence**: Machine Learning, Natural Language Processing, RASA, Prompt Engineering, PyTorch, TensorFlow, Predictive Modeling, Scikit-Learn, Deep Learning, Retrieval-Augmented Generation(RAG), Agentic AI, LangChain, LangGraph, Vector Databases
- **Data Science & Analytics**: Statistical Analysis, Data Visualization, Tableau, Power BI
- **Programming**: Python, Scala
- **Databases & Storage**: PostgreSQL, MySQL, PL/SQL, Oracle SQL, MongoDB, Redis, NoSQL Databases, Parquet, Avro, ORC
- **Backend, DevOps & Infrastructure**: REST APIs, FastAPI, Django, Microservices Architecture, ReactJS, Linux / Unix Shell Scripting, Docker, Terraform, CloudFormation, IaC, Git, GitHub, Jenkins, CI/CD Pipelines, ELK Stack, Monitoring & Observability

PROJECTS

Customer & Product Recommendation System **Feb 2025 - Jun 2025**
- Built hybrid recommendation pipelines using PySpark and Snowflake on AWS, improving recommendation relevance by 18–20% through collaborative filtering, NLP-based feature engineering, and optimized similarity scoring.
- Engineered user behavior features (RFM, recency signals) in Parquet and Delta Lake formats, improving customer segmentation accuracy by 18–20% through K-Means clustering, PCA, and dbt transformations.

Smart Blood Bank Recommendation System **Jan 2024 - May 2024**
- Built Airflow and PySpark-orchestrated data pipelines on AWS to ingest, clean, and integrate donor, inventory, and demand datasets in Parquet format, enabling scalable processing and reliable downstream analytics.
- Developed predictive matching and prioritization logic using Apache Spark across Blood availability and demand constraints, reducing blood supply risk by ~30% and supporting faster operational decisions.

Distributed Real-Time Chat & Broadcast System **Feb 2024 - May 2024**
- Built a distributed real-time chat and broadcast system on AWS using Kafka and WebSockets, supporting one-to-one and group messaging with sub-200ms end-to-end latency across multiple nodes under concurrent load.
- Implemented fault-tolerant message routing and workload distribution, improving system throughput and reliability by 30% during high-concurrency usage.

CERTIFICATIONS

- **Generative AI for Software Engineer: IBM, Mar 2026**
- **Google Analytics Professional Certificate: Google, Mar 2026**
- **AI Foundations for Everyone: IBM, Oct 2025**
- **Generative AI for Data Scientists: IBM, Oct. 2025**
- **Introduction to Artificial Intelligence (AI): Coursera, Jan 2021**